

1.

(a) Give the definition of the matrix exponent.

(b) Let $A_{[n \times n]}$, $B_{[n \times n]}$ be square matrices. Whether it true, that $e^{A+B} = e^A e^B$?

2. Give Poincare classification of singular points of a second-order linear homogeneous system with constant coefficients. You have to give all possible types of singular points, draw the pictures and specify the conditions.



ulate the theorem on the stability of the zero solution of the first approximation

the equation

$$y'' + 4y' + 4y = xe^{2x}.$$

5. Solve the equation



he equation

$$y'' + y = 2\left(\frac{1}{\cos x}\right)^3.$$



particular solution $y_1 = e^x - 1$ is given.

he equation

$$(e^x + 1)y'' - 2y' - e^x y = 0.$$

$$y''^3 + xy'' = 2y'.$$

8. Solve the equation

$$\begin{cases} \dot{x} = 2x - y \\ \dot{y} = x + 2e^t \end{cases}$$

9. Find all equilibria and study the stability

$$\begin{cases} \dot{x} = e^y - e^x \\ \dot{y} = \sqrt{3x + y^2} - 2 \end{cases}$$